

THEORETICAL PROBLEMS1. Prove $1 \Leftrightarrow 4$ of 2.3

• f is a function, Γ it's graph

1) f is convex

4) $\forall x_1, \dots, x_k \in D; \alpha_1, \dots, \alpha_k \in [0, 1], \sum_{i=1}^k \alpha_i = 1$:

$$f\left(\sum_{i=1}^k \alpha_i x_i\right) \leq \sum_{i=1}^k \alpha_i f(x_i)$$

• $1 \Leftrightarrow 4$ can be proved with induction.

• $f: D \rightarrow \mathbb{R}$ is convex if D is convex, and $\forall x, y \in D, \forall t \in [0, 1]$:

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y)$$

a) $1 \Rightarrow 4$

Base₁: $k=1 \rightarrow \alpha_1=1$

$$f(1 \cdot x_1) \leq 1 \cdot f(x_1) \quad \checkmark$$

Base₂: $k=2 \rightarrow \alpha_1 + \alpha_2 = 1 \rightarrow \alpha_2 = 1 - \alpha_1$

$$f(\alpha_1 x_1 + (1 - \alpha_1)x_2) \leq \alpha_1 f(x_1) + (1 - \alpha_1)f(x_2) \quad \checkmark$$

- holds, because we assume ① holds (f is convex)

Inductive step:

Assume statement holds for $k \geq (k \in \mathbb{N})$.

It must also hold for $k+1$

$$\bullet x_1, x_2, \dots, x_{k+1} \in \mathbb{D} \quad \bullet \alpha_1, \alpha_2, \dots, \alpha_{k+1} \in [0, 1], \sum_{i=1}^{k+1} \alpha_i = 1$$

$$f\left(\sum_{i=1}^{k+1} \alpha_i x_i\right) = f\left(\alpha_{k+1} x_{k+1} + \sum_{i=1}^k \alpha_i x_i\right) \Rightarrow \text{if } \alpha_{k+1} = 1, \text{ other } \alpha_i \text{ must be } 0$$

weights sum to 1:

$$= f\left(\alpha_{k+1} x_{k+1} + (1 - \alpha_{k+1}) \underbrace{\sum_{i=1}^k \frac{\alpha_i}{1 - \alpha_{k+1}} x_i}_{= y}\right) \Rightarrow \text{assume } \underline{\alpha_{k+1} < 1}$$
$$\alpha_{k+1} + (1 - \alpha_{k+1}) = 1 \checkmark$$

pull the weights out:

$$f\left(\alpha_{k+1} x_{k+1} + (1 - \alpha_{k+1}) y\right) \leq \alpha_{k+1} f(x_{k+1}) + (1 - \alpha_{k+1}) f(y) =$$

$$= \alpha_{k+1} f(x_{k+1}) + (1 - \alpha_{k+1}) f\left(\sum_{i=1}^k \frac{\alpha_i}{1 - \alpha_{k+1}} x_i\right)$$

$$\sum_{i=1}^k \frac{\alpha_i}{1 - \alpha_{k+1}} = \frac{1}{1 - \alpha_{k+1}} \sum_{i=1}^k \alpha_i \Rightarrow \sum_{i=1}^k \alpha_i = 1 - \alpha_{k+1}$$

BY I.H.:

$$f\left(\sum_{i=1}^k \frac{\alpha_i}{1 - \alpha_{k+1}} x_i\right) \leq \sum_{i=1}^k \frac{\alpha_i}{1 - \alpha_{k+1}} f(x_i)$$

$$\Rightarrow \leq \alpha_{k+1} f(x_{k+1}) + (1 - \alpha_{k+1}) \left(\sum_{i=1}^k \frac{\alpha_i}{1 - \alpha_{k+1}} f(x_i)\right) =$$

$$= \alpha_{k+1} f(x_{k+1}) + \sum_{i=1}^k \alpha_i f(x_i) = \sum_{i=1}^{k+1} \alpha_i f(x_i)$$

$\Rightarrow 4 \checkmark$

b) $4 \Rightarrow 1$

statement 4 holds for any $k \in \mathbb{N}$

prove f is convex

- choose $k=2$
- $x_1 = x, x_2 = y$
- $\alpha_1 = t, \alpha_2 = 1-t \quad (t \in [0,1])$
- $\alpha_1 + \alpha_2 = t + 1-t = 1 \quad \checkmark$

substitute into 4):

$$f(\alpha_1 x_1 + \alpha_2 x_2) \leq \alpha_1 f(x_1) + \alpha_2 f(x_2)$$

$$\underline{f(tx + (1-t)y) \leq tf(x) + (1-t)f(y)}$$

$4 \Rightarrow 1 \quad \checkmark$

$\square$

2. $f(x, y) = x^2 + e^x + y^2 - xy$

f restricted to $K = [-2, 2] \times [-2, 2]$ is Lipschitz, smooth, str. convex

- find (optimal) L, α, β on K
- prove f is convex

∇f :

$$\frac{\partial f}{\partial x} = 2x + e^x - y$$

$$\frac{\partial f}{\partial y} = 2y - x$$

$$\nabla f(x, y) = \begin{bmatrix} 2x + e^x - y \\ 2y - x \end{bmatrix}$$

$\nabla^2 f$:

$$\frac{\partial^2 f}{\partial x^2} = 2 + e^x$$

$$\frac{\partial^2 f}{\partial y^2} = 2$$

$$\frac{\partial^2 f}{\partial x \partial y} = -1 = \frac{\partial^2 f}{\partial y \partial x}$$

$$\nabla^2 f(x, y) = \begin{bmatrix} 2 + e^x & -1 \\ -1 & 2 \end{bmatrix}$$

a) f is convex

- if f is twice cont. diff. it is convex if all eigenvals. of $\nabla^2 f$ are non-neg. ($\nabla^2 f$ is PSD)

$$\nabla^2 f(x, y) = \begin{bmatrix} 2 + e^x & -1 \\ -1 & 2 \end{bmatrix}$$

Sylvester:

$$\det A_1 = 2 + e^x \rightarrow e^x > 0$$

$$\det A_1 > 0$$

$\nabla^2 f(x, y)$ is P.D.,

$$\det A_2 = (2 + e^x) \cdot 2 - 1 = 3 + 2e^x$$

therefore f is convex.

$$\det A_2 > 0$$

b) Find α and β .

- f is β -smooth, if all eigenvalues of $\nabla^2 f$ lie on $[0, \beta]$
- f is α -str. conv., if all eigenvalues of $\nabla^2 f$ are $\geq \alpha$

$$\det(\nabla^2 f - \lambda I) = 0$$

$$(2+e^x - \lambda)(2-\lambda) - 1 = 0$$

$$\lambda^2 - (4+e^x)\lambda + (3+2e^x) = 0$$

$$\lambda_{1,2} = \frac{4+e^x \pm \sqrt{(4+e^x)^2 - 4(3+2e^x)}}{2} \rightarrow \begin{aligned} &16 + 8e^x + e^{2x} - 12 - 8e^x \\ &= 4 + e^{2x} \end{aligned}$$

$$\lambda_{1,2} = \frac{4+e^x \pm \sqrt{4+e^{2x}}}{2}$$

$$\begin{aligned} \lambda_1(x) &= \frac{4+e^x - \sqrt{4+e^{2x}}}{2} \\ \lambda_2(x) &= \frac{4+e^x + \sqrt{4+e^{2x}}}{2} \end{aligned}$$

$$x \in [-2, 2]$$

min of $\lambda_1(x)$

α : e^x grows faster than $\sqrt{4+e^{2x}} \rightarrow \lambda_1(x)$ increases with x
 $\Rightarrow$ take $x = -2$

$$\alpha = \lambda_1(-2) = \frac{4+e^{-2} - \sqrt{4+e^{-4}}}{2}$$

max of $\lambda_2(x)$

β : $\lambda_2(x)$ increases with $x \Rightarrow$ take $x = 2$

$$\beta = \lambda_2(2) = \frac{4+e^2 + \sqrt{4+e^4}}{2}$$

c) Find L

• f is L -Lip. iff $\|\nabla f\| \leq L$

→ find max. of $\|\nabla f(x,y)\|^2$ on D

$$\|\nabla f(x,y)\|^2 = (2x + e^x - y)^2 + (2y - x)^2$$

• we have a convex func. on bounded square domain.
⇒ max. must be on one of the corners.

1) $(2,2)$: $(4 + e^2 - 2)^2 + (4 - 2)^2 = (2 + e^2)^2 + 4$

2) $(-2,2)$: $(-4 + e^{-2} - 2)^2 + (4 - (-2))^2 = (-6 + e^{-2})^2 + 36$

3) $(-2,-2)$: $(-4 + e^2 - (-2))^2 + (-4 - (-2))^2 = (-2 + e^2)^2 + 4$

4) $(2,-2)$: $(4 + e^2 - (-2))^2 + (-4 - 2)^2 = (6 + e^2)^2 + 36$

↓
clearly the largest

$$L = \sqrt{(6 + e^2)^2 + 36}$$

$$\rightarrow 36 + 12e^2 + e^4 + 36 = 72 + 12e^2 + e^4$$

$$L = \sqrt{72 + 12e^2 + e^4}$$

3.1 $\mathbb{R}^2 \rightarrow \mathbb{K}$

$$x^2 + y^2 \leq 1.5$$

$$[-1, 1] \times [-1, 1]$$

$$\Delta: (-1, -1), (1.5, -1), (-1, 1.5)$$

Find formulas π_K

1) DISK: $x^2 + y^2 \leq 1.5$

RADIAL PROJECTION: • closest point lies on the boundary
→ take point's vector and shrink to $\sqrt{1.5}$

$$v = (x, y), \quad \|v\| = \sqrt{x^2 + y^2}$$

$$\pi_K(x, y) = \begin{cases} (x, y) & \text{if } x^2 + y^2 \leq 1.5 \\ \frac{\sqrt{1.5}}{\sqrt{x^2 + y^2}} (x, y) & \text{if } x^2 + y^2 > 1.5 \end{cases}$$

2) SQUARE: $[-1, 1] \times [-1, 1]$

• x and y act independently. Project x to $[-1, 1]$ and y to $[-1, 1]$

$$\pi_K(x, y) = (\max(-1, \min(1, x)), \max(-1, \min(1, y)))$$

3) TRIANGLE: $(-1, -1), (1.5, -1), (-1, 1.5)$



slope of hypotenuse: $m = \frac{-1 - 1.5}{1.5 + 1} = \frac{-2.5}{2.5}$

$$m = -1$$

point slope: $y - y_1 = m(x - x_1) \rightarrow y - 1.5 = -1(x + 1)$

$$y - 1.5 = -x - 1$$

$$y = \cancel{0.5} = x$$

$$\underline{y = -x + 0.5} \longrightarrow y + x = 0.5$$

⇓

• a point will project to:

- one of the edges
- one of the corners
+ itself

K is defined:

$$\begin{cases} x \geq -1 \\ y \geq -1 \\ x + y \leq 0.5 \end{cases}$$

7 (3+3+1)

1) INSIDE: $x \geq -1, y \geq -1, x + y \leq 0.5$

• maps to (x, y)

2) BOTTOM L. COR: $y < -1, x < -1$

• to $(-1, -1)$

3) BOTTOM EDGE: ~~$y < -1$~~ $y < -1, -1 \leq x \leq 1.5$

• to $(x, -1)$

4) LEFT EDGE: $x < -1, -1 \leq y \leq 1.5$

• to $(-1, y)$

5) BOTTOM R. COR: $x > 1.5, y < -1$

• to $(1.5, -1)$ below normal line

6) TOP L. COR: $y > 1.5, x < -1$

• to $(-1, 1.5)$ left of normal line

7) HYPOTHENUSE: $x + y > 0.5$, between 5) and 6)

- use the normal vector

• to $\left(\frac{x - y + 0.5}{2}, \frac{y - x + 0.5}{2} \right)$

$$\begin{cases} y - 1.5 = 1(x - (-1)) \\ y = x + 2.5 \end{cases} \quad 6$$

$$\begin{cases} y - (-1) = 1(x - 1.5) \\ y = x - 2.5 \end{cases} \quad 5$$



~~π_k~~

$$\pi_k(x, y) = \begin{cases} (x, y) & \text{if } x \geq -1, y \geq -1, x+y \leq 0.5 \\ (-1, -1) & \text{if } x < -1, y < -1 \\ (x, -1) & \text{if } -1 \leq x \leq 1.5, \text{ ~~and~~ } y < -1 \\ (-1, y) & \text{if } x < -1, -1 \leq y \leq 1.5 \\ (1.5, -1) & \text{if } x - y > 2.5, y < x - 2.5 \text{ (5)} \\ (-1, 1.5) & \text{if } y - x > 2.5, y > \text{~~x~~ } x + 2.5 \text{ (6)} \\ \left(\frac{x-y+0.5}{2}, \frac{y-x+0.5}{2} \right) & \text{if } x+y > 0.5 \text{ and between (5) and (6)} \end{cases}$$

$$4.1 \quad f(x, y) = x^2 + 2y^2$$

$$x_1 = (1, 1)$$

$$a) \text{ min of } f(x_2)$$

$$b) \text{ min } \|x_2 - x^*\|$$

$$x_{k+1} = x_k - \gamma \nabla f(x_k), \quad \gamma > 0$$

$$\nabla f(x, y) = \begin{bmatrix} 2x \\ 4y \end{bmatrix} \quad \nabla f(1, 1) = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

$$x_2(\gamma) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \gamma \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 - 2\gamma \\ 1 - 4\gamma \end{bmatrix}$$

$$\begin{aligned} a) \quad f(x_2(\gamma)) &= (1 - 2\gamma)^2 + 2(1 - 4\gamma)^2 \\ &= \cancel{1} - 4\gamma + 4\gamma^2 + \cancel{2} - 16\gamma + 32\gamma^2 \\ &= 36\gamma^2 - 20\gamma + 3 = F \end{aligned}$$

Find optimal γ :

$$\frac{\partial F}{\partial \gamma} = 72\gamma - 20$$

$$72\gamma = 20$$

$$\gamma = \frac{20}{72} = \frac{10}{36} = \underline{\underline{\frac{5}{18}}}$$

$$\begin{aligned} f(x_2(\frac{5}{18})) &= 36(\frac{5}{18})^2 - 20(\frac{5}{18}) + 3 = 36(\frac{25}{324}) - \frac{100}{18} + 3 \\ &= \frac{25}{9} - \frac{50}{9} + \frac{27}{9} = \boxed{\frac{2}{9}} \text{ min. func. value achievable in 1 step is } 2/9. \end{aligned}$$

$$b) x^* = (0, 0) \text{ (trivially)}$$

$$\text{minimize } \|x_2 - x^*\|$$

$$D(\gamma) = \|x_2(\gamma) - x^*\|^2 = (1 - 2\gamma - 0)^2 + (1 - 4\gamma - 0)^2$$

$$D(\gamma) = 1 - 4\gamma + 4\gamma^2 + 1 - 8\gamma + 16\gamma^2 = 20\gamma^2 - 12\gamma + 2$$

$$\frac{\partial D}{\partial \gamma} = 40\gamma - 12$$

$$40\gamma = 12$$

$$\gamma = \frac{12}{40} = \underline{\underline{\frac{3}{10}}}$$

$$D\left(\frac{3}{10}\right) = 20\left(\frac{3}{10}\right)^2 - 12\left(\frac{3}{10}\right) + 2 = 20\left(\frac{9}{100}\right) - \frac{36}{10} + 2$$

$$= \frac{18}{10} - \frac{36}{10} + \frac{20}{10} = \frac{2}{10} = \frac{1}{5}$$

$$\Downarrow \\ \sqrt{\frac{1}{5}} = \frac{\sqrt{5}}{5}$$

The closest we can get to the minimum in 1 step is $\frac{\sqrt{5}}{5}$.